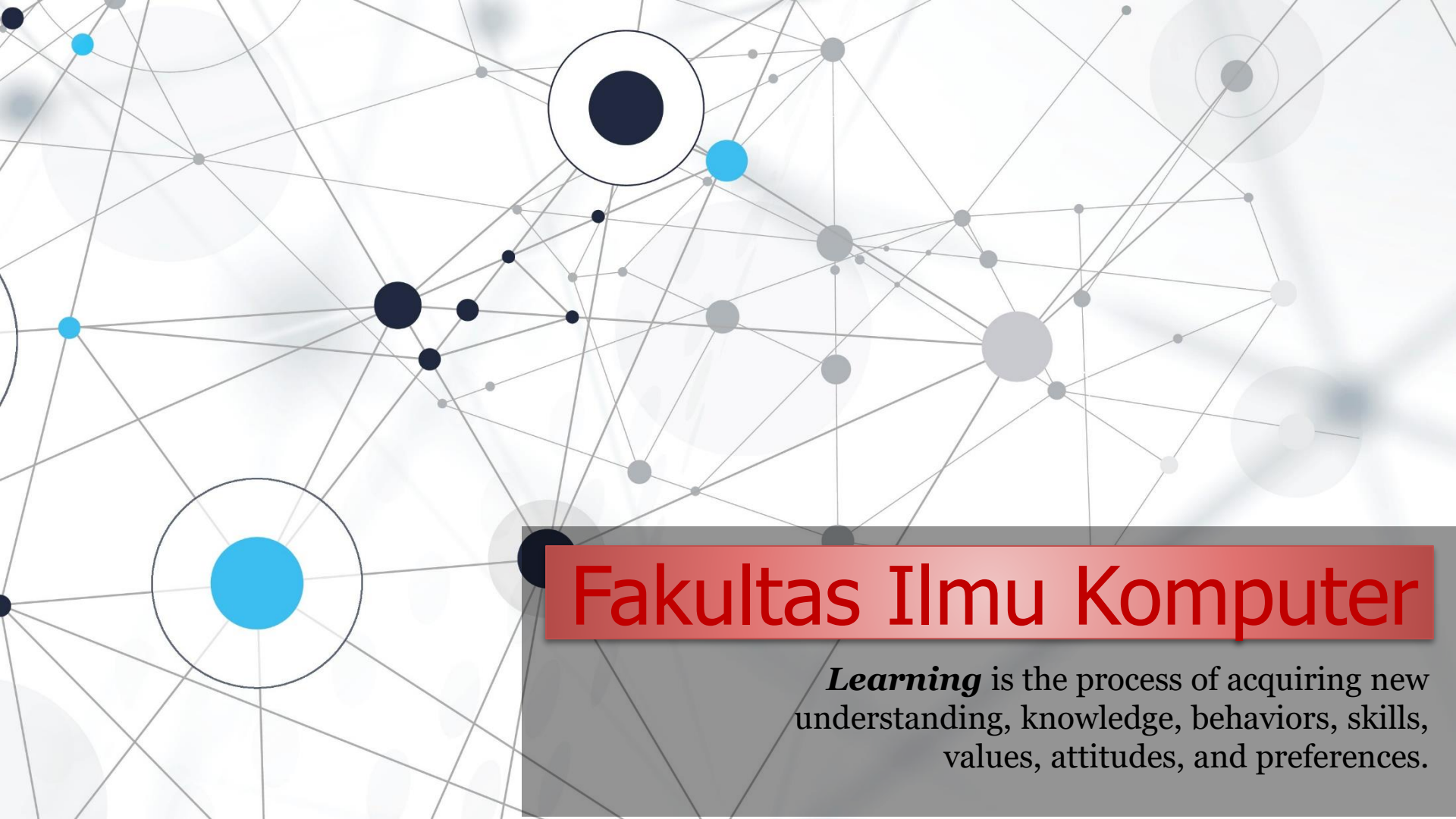




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Framework #7-1

[Create Database & Tables]



Welcome to the Framework Programming *course*

Three Important MVC Components (*layers*)

Component #1 → Model :

- ✓ The model component stores data and its related logic. It represents data that is being transferred between controller components or any other related business logic. For example, a Controller object will retrieve the customer info from the database. It manipulates data and send back to the database or use it to render the same data.
- ✓ It responds to the request from the views and also responds to instructions from the controller to update itself. It is also the lowest level of the pattern which is responsible for maintaining data.

Component #2 → View :

- ✓ The View is that part of the application that represents the presentation of data.
- ✓ Views are created by the data collected from the model data. A view requests the model to give information so that it resents the output presentation to the user.
- ✓ The view also represents the data from chats, diagrams, and table. For example, any customer view will include all the UI components like text boxes, drop downs, etc.

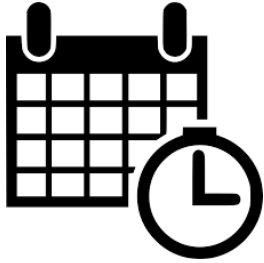
Component #3 → Controller :

- ✓ The Controller is that part of the application that handles the user interaction. The controller interprets the mouse and keyboard inputs from the user, informing model and the view to change as appropriate.
- ✓ A Controller send's commands to the model to update its state (*E.g., Saving a specific document*). The controller also sends commands to its associated view to change the view's presentation (*For example scrolling a particular document*).

The paper was published under the title:

Burbeck, Steve. "Applications programming in smalltalk-80 (*tm*): How to use model-view-controller (mvc)." *Smalltalk-80 v2 5* (1992): 1-11.

Three Important MVC Components (*layers*)



Model: It includes all the data and its related logic.



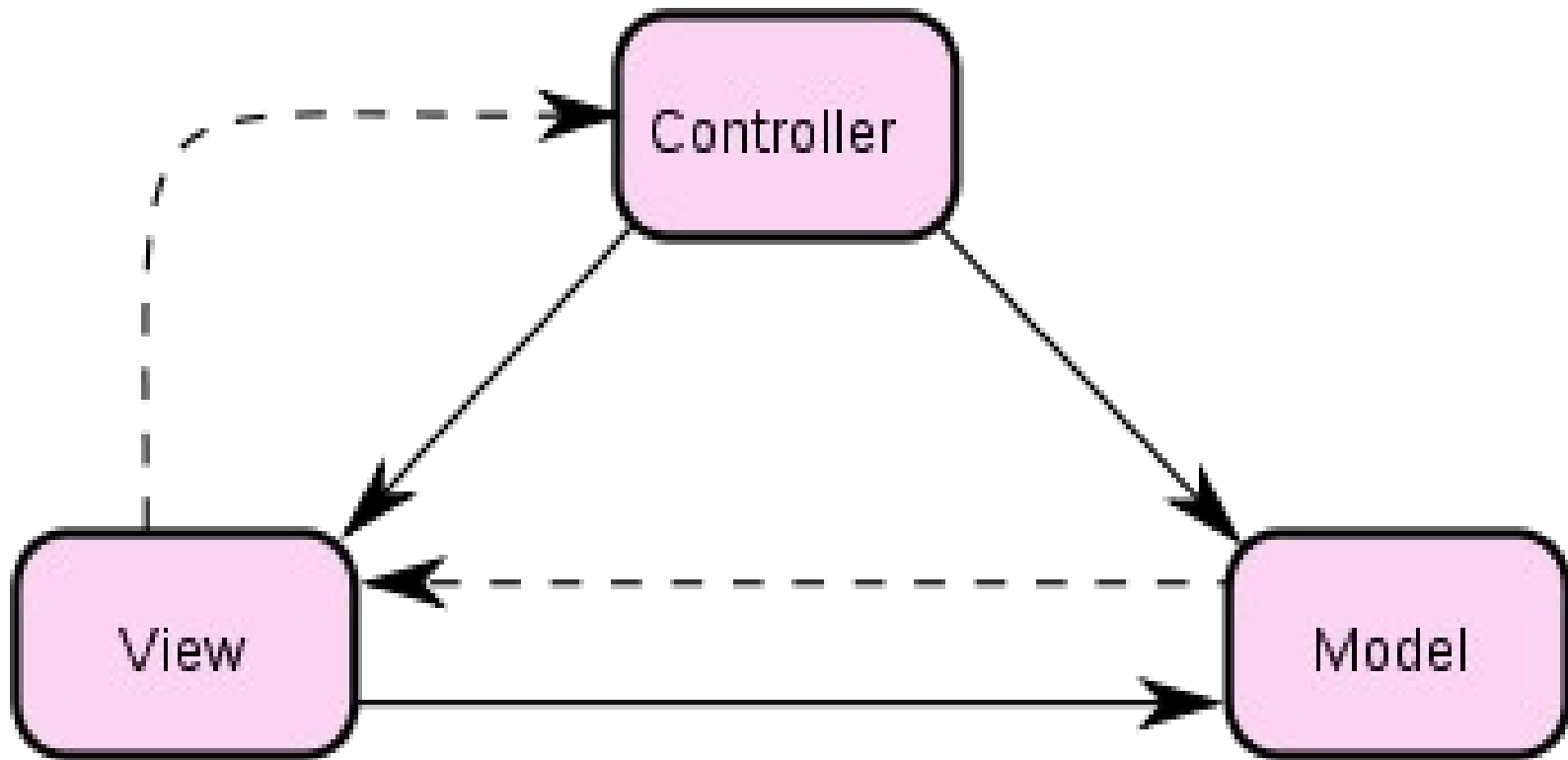
View: Present data to the user or handles user interaction.



Controller: An interface between Model and View components.

Three Important MVC Components (*layers*)

logic program (*controller*)



Tampilan (*view*) => UI

Manage the data

phpMyAdmin in XAMPP

Bringing MySQL to the web

phpMyAdmin is a free software tool written in PHP, intended to handle the administration of MySQL over the Web. phpMyAdmin supports a wide range of operations on MySQL and MariaDB. Frequently used operations (***managing databases, tables, columns, relations, indexes, users, permissions, etc***) can be performed via the user interface, while you still have the ability to directly execute any SQL statement.



Read Source:

<https://www.phpmyadmin.net/>

MySQL

MySQL is a relational database management system based on **SQL – Structured Query Language**. The application is used for a wide range of purposes, including data warehousing, e-commerce, and logging applications. The most common use for mySQL however, is for the purpose of a web database. It can be used to store anything from a single record of information to an entire inventory of available products for an online store. In association with a scripting language such as **PHP** or **Perl**, it is possible to create websites which will interact in real-time with a mySQL database to rapidly display categorised and searchable information to a website user.



Read Source:

<https://www.123-reg.co.uk/support/servers/what-is-mysql-and-why-do-i-need-it/>

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1 – Create database in XAMPP using phpMyAdmin

A decorative border of green leaves and branches at the bottom of the slide.



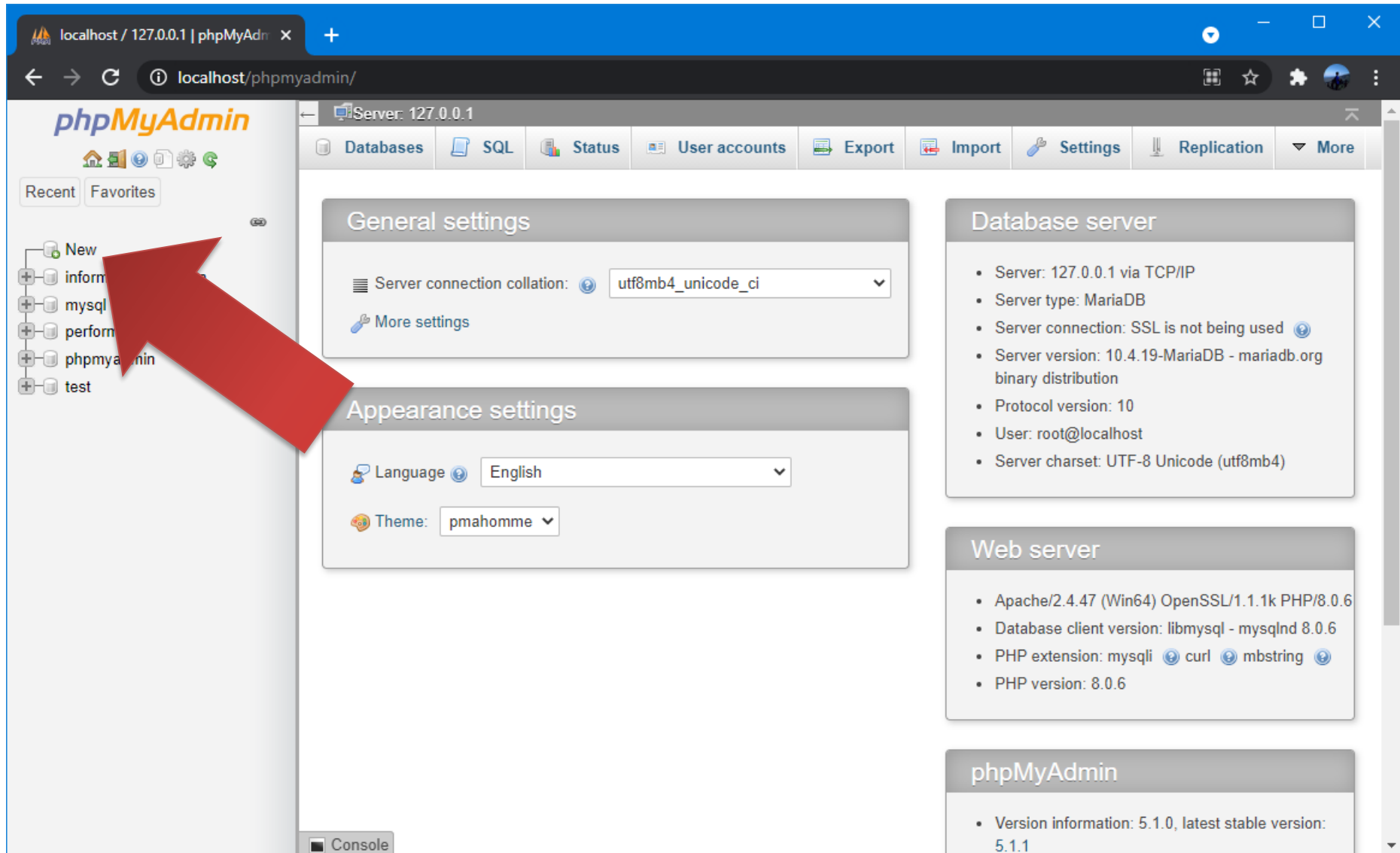
XAMPP phpMyAdmin

How to Create Database and Tables

Recommended to install XAMPP before we start to this lecture. We might need to test our AIMVC project can run locally. If you find any problems related to installing and configuring XAMPP phpMyAdmin, then please find previous lecture in this class or any tutorials to solve it.

Navigate to Database on phpMyAdmin

Navigate cursor to <http://localhost/phpmyadmin/> and click button “new”.



The screenshot displays the phpMyAdmin web interface in a browser window. The address bar shows `localhost / 127.0.0.1 | phpMyAdmin` and the URL `localhost/phpmyadmin/`. The interface includes a top navigation bar with tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, and More. A left sidebar contains a tree view of databases: New, inform, mysql, perform, phpmyadmin, and test. A red arrow points to the 'New' button in this sidebar. The main content area is divided into several panels: General settings (Server connection collation: utf8mb4_unicode_ci), Appearance settings (Language: English, Theme: pmahomme), Database server (Server: 127.0.0.1 via TCP/IP, Server type: MariaDB, Server connection: SSL is not being used, Server version: 10.4.19-MariaDB - mariadb.org binary distribution, Protocol version: 10, User: root@localhost, Server charset: UTF-8 Unicode (utf8mb4)), Web server (Apache/2.4.47 (Win64) OpenSSL/1.1.1k PHP/8.0.6, Database client version: libmysql - mysqlnd 8.0.6, PHP extension: mysqli, curl, mbstring, PHP version: 8.0.6), and phpMyAdmin (Version information: 5.1.0, latest stable version: 5.1.1). A Console tab is visible at the bottom left.

Create a Database

The screenshot shows the phpMyAdmin web interface. The browser address bar indicates the URL is `localhost/phpmyadmin/index.php?route=/server/databases`. The interface has a top navigation bar with tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, and More. The left sidebar shows a tree view of databases: information_schema, mysql, performance_schema, phpmyadmin, and test. The main content area is titled 'Databases' and features a 'Create database' section. In this section, the input field for the database name contains 'db_university', and the collation dropdown is set to 'utf8mb4_general_ci'. A 'Create' button is positioned to the right of the dropdown. Below the 'Create database' section, there is a table listing existing databases:

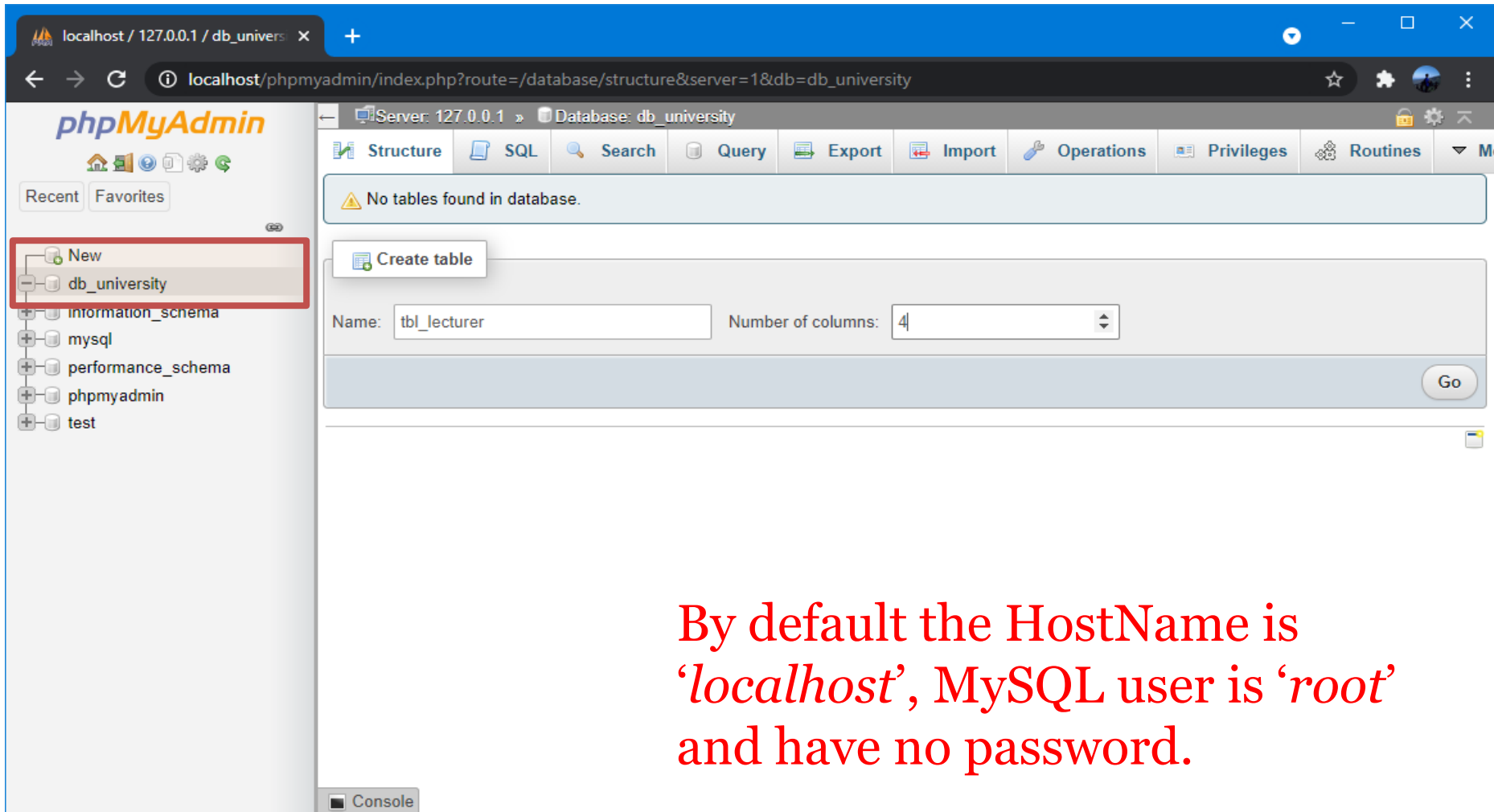
Database	Collation	Action
☐ information_schema	utf8_general_ci	Check privileges
☐ mysql	utf8mb4_general_ci	Check privileges
☐ performance_schema	utf8_general_ci	Check privileges
☐ phpmyadmin	utf8_bin	Check privileges
☐ test	latin1_swedish_ci	Check privileges

Total: 5

Below the table, there are checkboxes for 'Check all' and a 'Drop' button. A note at the bottom states: 'Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.' with a link to 'Enable statistics'.

See the option to “*Create a Database*” and input field to enter the database name. Write the database name and click the “*Create*” button.

Create Tables Using phpMyAdmin



By default the HostName is 'localhost', MySQL user is 'root' and have no password.

In the phpMyAdmin click on the “Structure” tab and click on the “Create new table on database” wizard to get started. You can see above fields, you can add the **table name** and the total number of fields (*columns*) and click the “Go” button.

Specify Data Type

localhost / 127.0.0.1 / db_univers

localhost/phpmyadmin/index.php?route=/database/structure&server=1&db=db_university

Server: 127.0.0.1 » Database: db_university » Table: tbl_lecturer

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Table name: tbl_lecturer Add 1 column(s) Go

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
id	VARCHAR	100	None			☐	---
Pick from Central Columns							
name	VARCHAR	100	None			☐	---
Pick from Central Columns							
email	VARCHAR	100	None			☐	---
Pick from Central Columns							
phone	VARCHAR	100	None			☐	---

Structure

Table comments: Collation: Storage Engine: InnoDB Connection:

PARTITION definition:

Partition by: (Expression or column list)

Partitions:

Preview SQL Save

Console

After the data type, you can specify other optional attributes for each *column*.

Table Structure

The screenshot shows the phpMyAdmin interface for a MySQL database named 'db_university'. The 'Table structure' tab is selected for the table 'tbl_lecturer'. The table has four columns: 'id', 'name', 'email', and 'phone', all of type 'varchar(100)' and collation 'utf8mb4_general_ci'. Below the table structure, there are options to 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'. There are also buttons to 'Add to central columns' and 'Remove from central columns'. At the bottom, there are sections for 'Indexes' and 'Partitions', both showing 'No index defined!' and 'No partitioning defined!' respectively. The 'Console' tab is visible at the bottom left.

localhost / 127.0.0.1 / db_univers x +

localhost/phpmyadmin/index.php?route=/table/structure&db=db_university&table=tbl_lecturer

Server: 127.0.0.1 » Database: db_university » Table: tbl_lecturer

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Table structure Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	2	name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	3	email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	4	phone	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)

[Add to central columns](#) [Remove from central columns](#)

[Print](#) [Propose table structure](#) [Track table](#) [Move columns](#) [Normalize](#)

[Add](#) 1 column(s) after phone [Go](#)

Indexes [?](#)

[No index defined!](#)

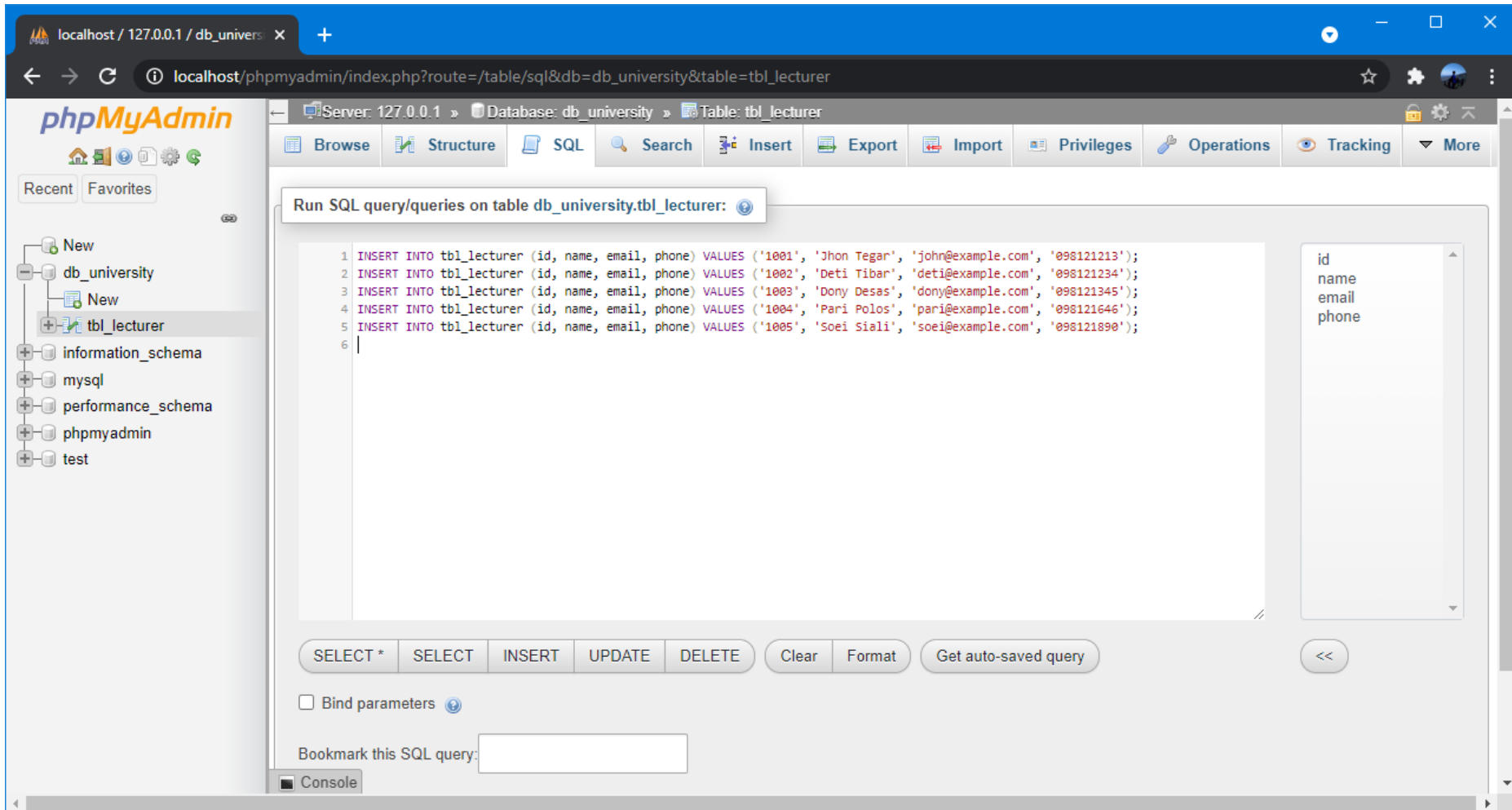
Create an index on 1 columns [Go](#)

Partitions [?](#)

[No partitioning defined!](#)

Console

MySQL Insert Data *via* SQL Suery



The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/table/sql&db=db_university&table=tbl_lecturer`. The interface shows the 'db_university' database selected, with the 'tbl_lecturer' table chosen. The 'SQL' tab is active, displaying a query to insert five rows of data into the 'tbl_lecturer' table. The query is as follows:

```
1 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1001', 'Jhon Tegar', 'john@example.com', '098121213');
2 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1002', 'Deti Tibar', 'deti@example.com', '098121234');
3 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1003', 'Dony Desas', 'dony@example.com', '098121345');
4 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1004', 'Pari Polos', 'pari@example.com', '098121646');
5 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1005', 'Soei Siali', 'soei@example.com', '098121890');
6
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. The 'INSERT' button is highlighted. There is also a checkbox for 'Bind parameters' and a 'Bookmark this SQL query' field.

```
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1001', 'Jhon Tegar', 'john@example.com', '098121213');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1002', 'Deti Tibar', 'deti@example.com', '098121234');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1003', 'Dony Desas', 'dony@example.com', '098121345');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1004', 'Pari Polos', 'pari@example.com', '098121646');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1005', 'Soei Siali', 'soei@example.com', '098121890');
```


Data from a MySQL Database

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/sql&db=db_university&table=tbl_lecturer&pos=0`. The interface is for the `db_university` database, specifically viewing the `tbl_lecturer` table. A message at the top states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available." Below this, a green status bar indicates "Showing rows 0 - 4 (5 total, Query took 0.0009 seconds.)". The SQL query entered is `SELECT * FROM `tbl_lecturer``. Below the query, there are options for "Profiling" and links for "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". A control bar shows "Show all", "Number of rows: 25", and a "Filter rows" search box. The query results are displayed in a table with the following data:

id	name	email	phone
1001	Jhon Tegar	john@example.com	098121213
1002	Deti Tibar	deti@example.com	098121234
1003	Dony Desas	dony@example.com	098121345
1004	Pari Polos	pari@example.com	098121646
1005	Soei Siali	soei@example.com	098121890

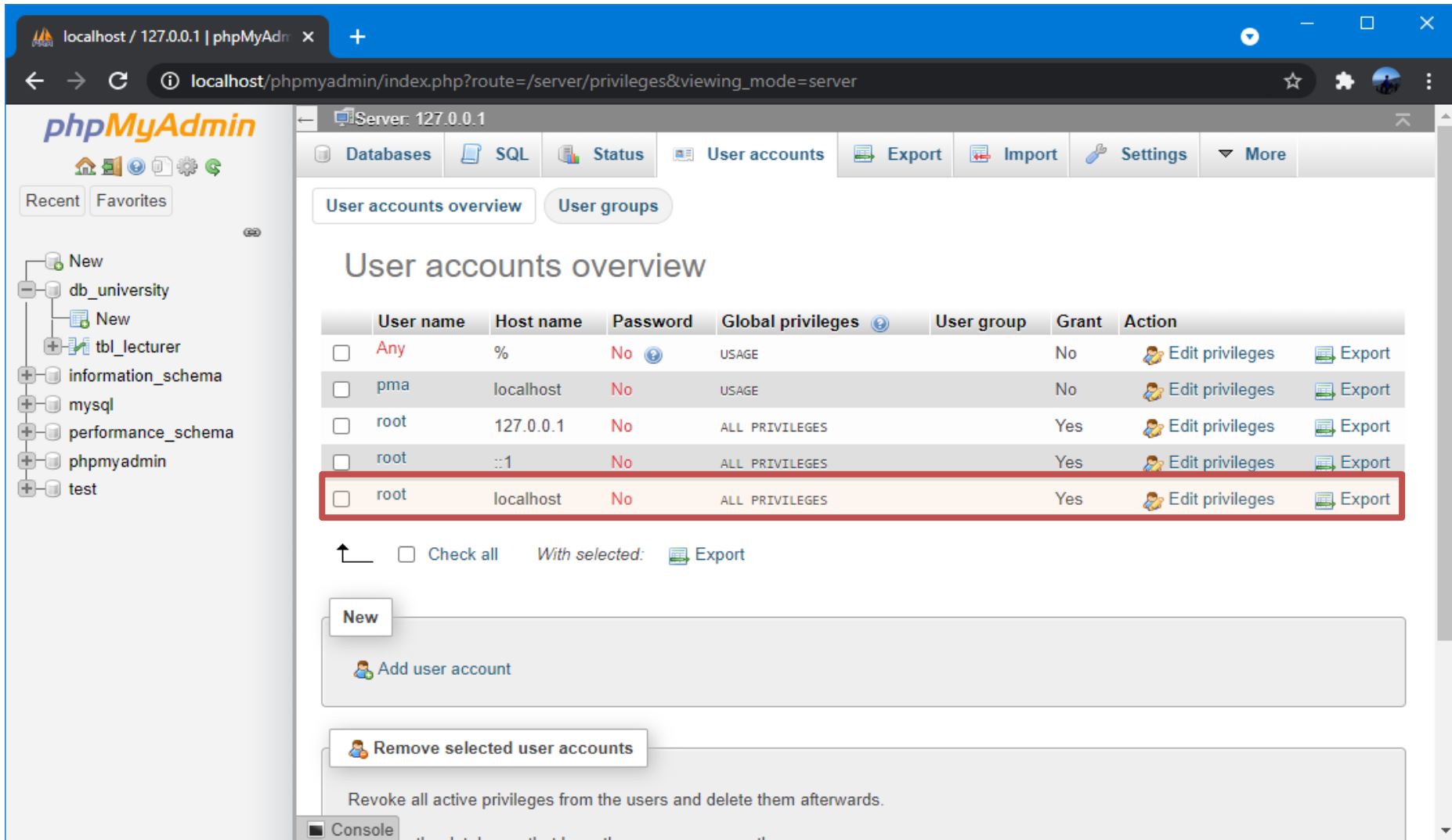
Below the table, there is another control bar with "Show all", "Number of rows: 25", and a "Filter rows" search box. At the bottom, there is a "Query results operations" section with buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view". The left sidebar shows the database structure with `db_university` selected and `tbl_lecturer` highlighted.

A decorative border of green leaves and branches at the top of the page.

2 – Set Password to phpMyAdmin *on* XAMPP

A decorative border of green leaves and branches at the bottom of the page.

User Accounts



The screenshot shows the phpMyAdmin interface for managing user accounts. The left sidebar contains a tree view of databases and schemas, including 'db_university', 'tbl_lecturer', 'information_schema', 'mysql', 'performance_schema', 'phpmyadmin', and 'test'. The main content area is titled 'User accounts overview' and displays a table of user accounts. The table has columns for 'User name', 'Host name', 'Password', 'Global privileges', 'User group', 'Grant', and 'Action'. The 'root' user with host 'localhost' is highlighted with a red border. Below the table, there are buttons for 'New' (Add user account) and 'Remove selected user accounts' (Revoke all active privileges from the users and delete them afterwards).

	User name	Host name	Password	Global privileges	User group	Grant	Action
☐	Any	%	No	USAGE		No	Edit privileges Export
☐	pma	localhost	No	USAGE		No	Edit privileges Export
☐	root	127.0.0.1	No	ALL PRIVILEGES		Yes	Edit privileges Export
☐	root	:::1	No	ALL PRIVILEGES		Yes	Edit privileges Export
☐	root	localhost	No	ALL PRIVILEGES		Yes	Edit privileges Export

☐ Check all With selected: [Export](#)

New

[Add user account](#)

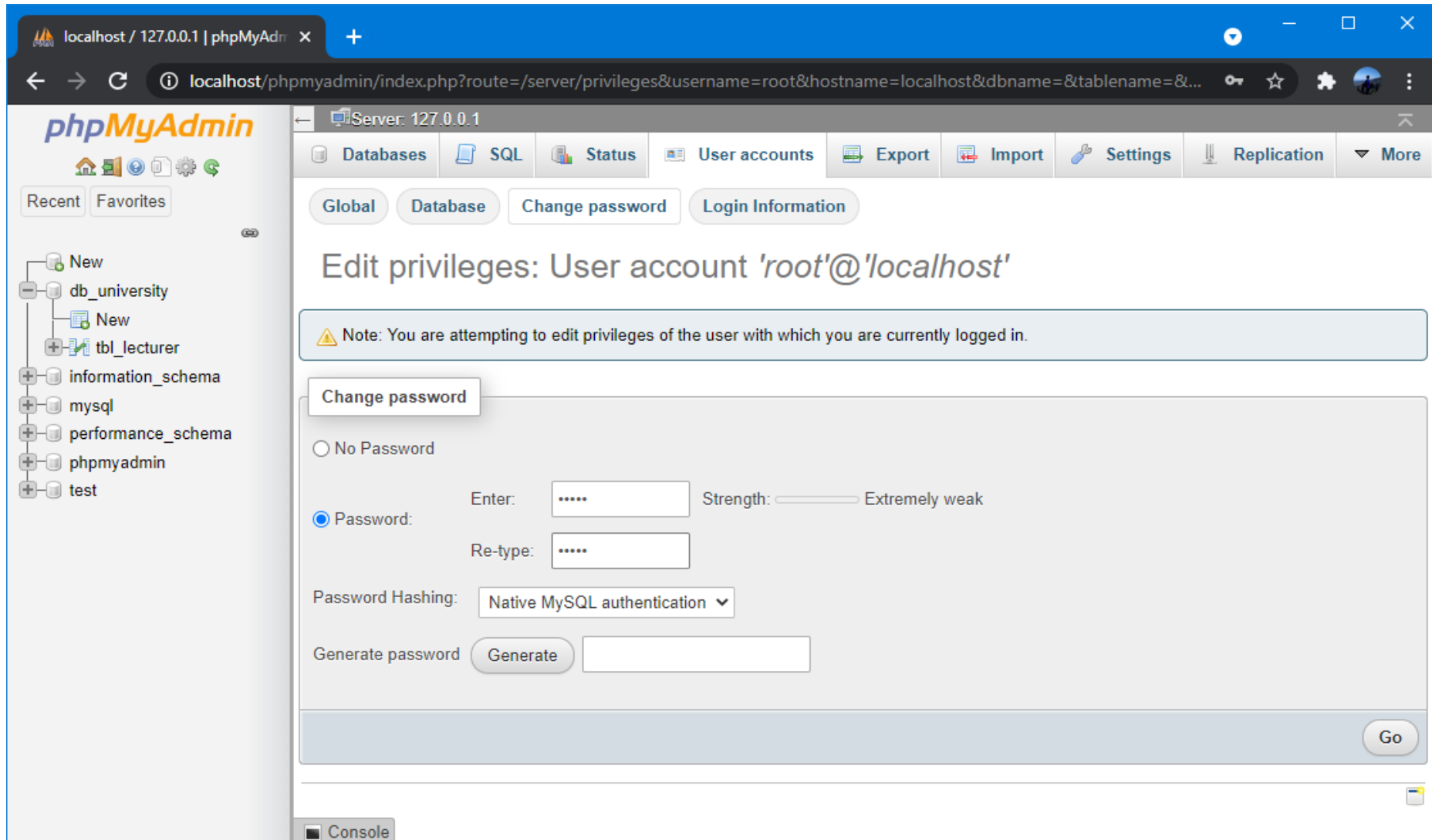
Remove selected user accounts

Revoke all active privileges from the users and delete them afterwards.

Console

Click on the “user accounts” option at the top of the page and press the “*edit privileges*” under “actions” option for the Username “root” and Hostname “localhost”.

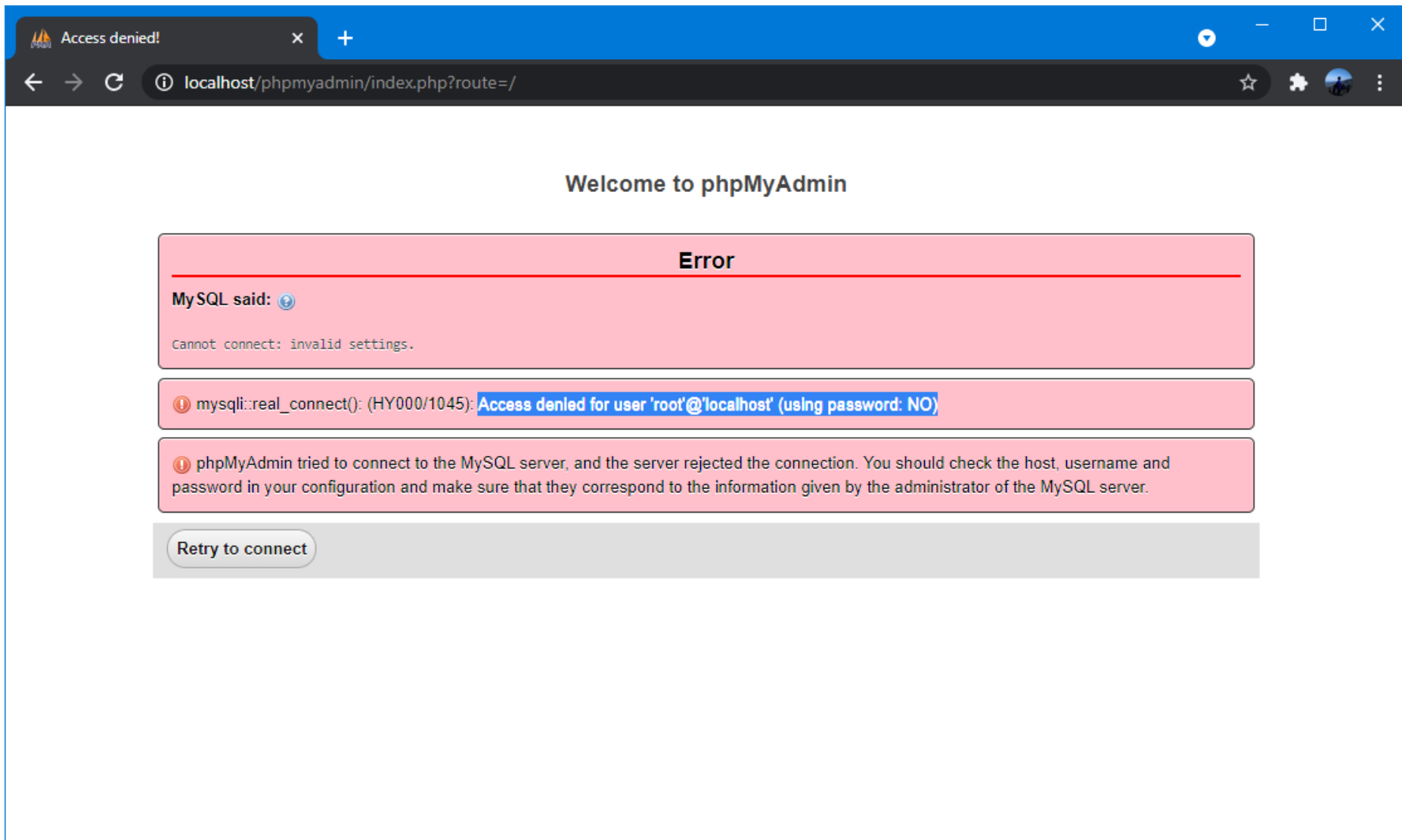
Change/Edit Password



The screenshot shows the phpMyAdmin web interface in a browser window. The address bar displays the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&username=root&hostname=localhost&dbname=&tablename=&...`. The interface has a sidebar on the left with a tree view of databases and schemas, including `db_university`, `information_schema`, `mysql`, `performance_schema`, `phpmyadmin`, and `test`. The main content area has a top navigation bar with tabs: `Databases`, `SQL`, `Status`, `User accounts`, `Export`, `Import`, `Settings`, `Replication`, and `More`. Below this, there are sub-tabs: `Global`, `Database`, `Change password` (which is selected), and `Login Information`. The main heading is `Edit privileges: User account 'root'@'localhost'`. A yellow warning box states: `Note: You are attempting to edit privileges of the user with which you are currently logged in.` The `Change password` section contains a radio button for `No Password` and a selected radio button for `Password:`. The `Password:` section has an `Enter:` field with masked characters, a `Re-type:` field with masked characters, and a `Strength:` indicator showing `Extremely weak`. Below this is a `Password Hashing:` dropdown menu set to `Native MySQL authentication`. At the bottom of the form is a `Generate password` button and a `Generate` button next to an empty text field. A `Go` button is located at the bottom right of the form area. A `Console` tab is visible at the bottom left of the interface.

Choose the third tab “*change password*” and type your password in the provided field, retype the password to confirm it and then finally click on the “Go” button to conclude the process.

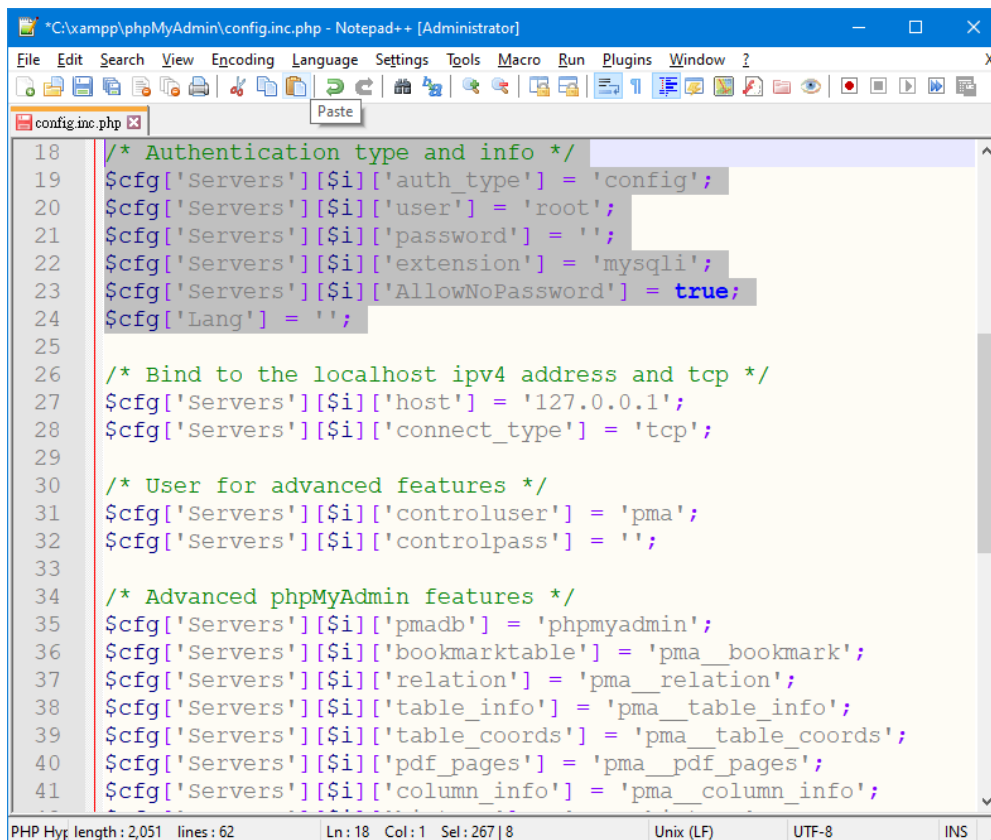
Error Access Denied



Configuration Settings [*allow without password*]

ERROR #1045 - Access denied for user 'root'@'localhost' (using password: NO) when opening http://localhost/phpmyadmin.

- ❖ Navigate to "C:\xampp\phpMyAdmin"
- ❖ Find the file "config.inc.php" and open in a Notepad or better, Notepad++ (**run as administrator**).
- ❖ At line # 18 to 24, is the settings for user authentication.



The screenshot shows a Notepad++ window titled "C:\xampp\phpMyAdmin\config.inc.php - Notepad++ [Administrator]". The file content is as follows:

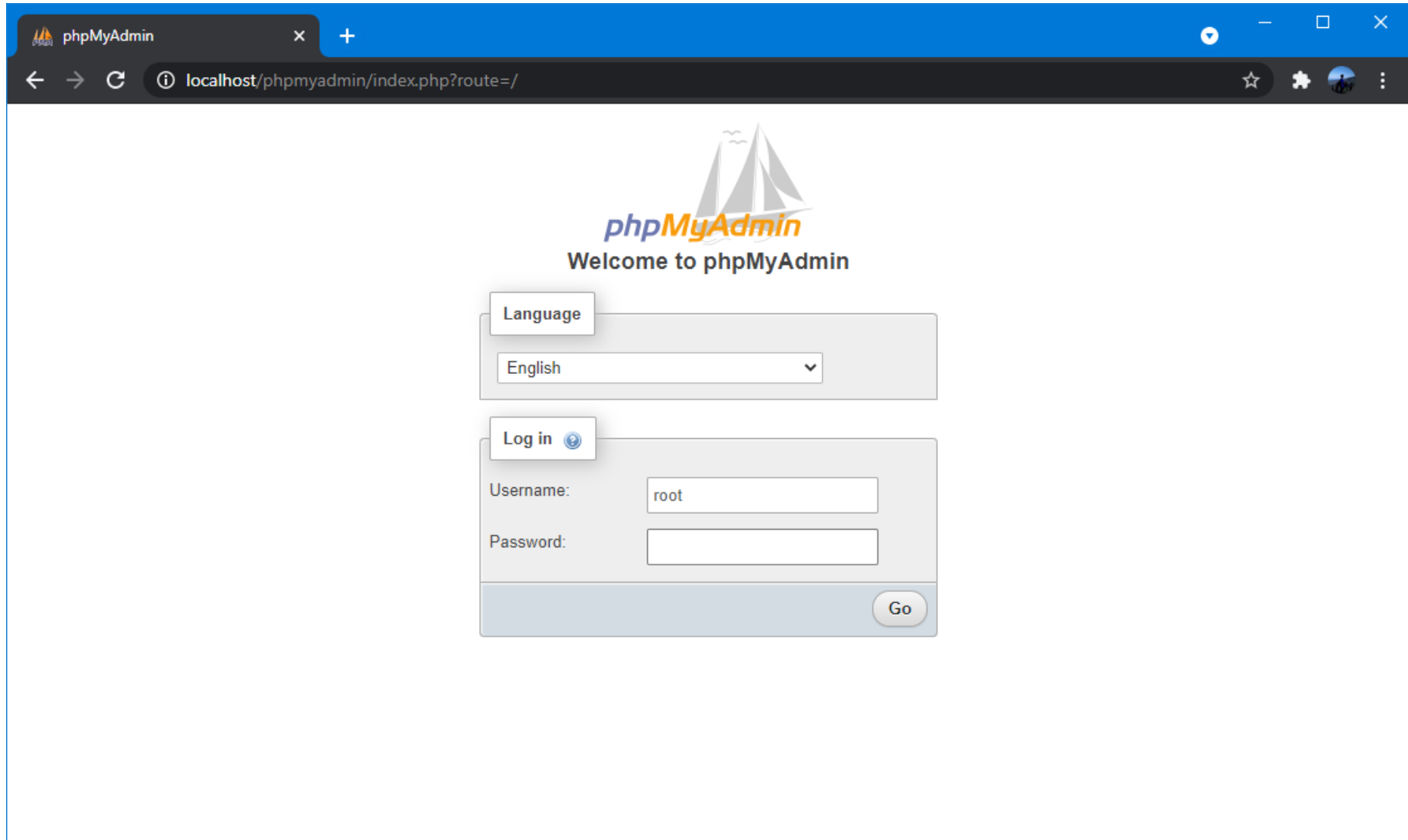
```
18  /* Authentication type and info */
19  $cfg['Servers'][$i]['auth_type'] = 'config';
20  $cfg['Servers'][$i]['user'] = 'root';
21  $cfg['Servers'][$i]['password'] = '';
22  $cfg['Servers'][$i]['extension'] = 'mysqli';
23  $cfg['Servers'][$i]['AllowNoPassword'] = true;
24  $cfg['Lang'] = '';
25
26  /* Bind to the localhost ipv4 address and tcp */
27  $cfg['Servers'][$i]['host'] = '127.0.0.1';
28  $cfg['Servers'][$i]['connect_type'] = 'tcp';
29
30  /* User for advanced features */
31  $cfg['Servers'][$i]['controluser'] = 'pma';
32  $cfg['Servers'][$i]['controlpass'] = '';
33
34  /* Advanced phpMyAdmin features */
35  $cfg['Servers'][$i]['pmadb'] = 'phpmyadmin';
36  $cfg['Servers'][$i]['bookmarktable'] = 'pma_bookmark';
37  $cfg['Servers'][$i]['relation'] = 'pma_relation';
38  $cfg['Servers'][$i]['table_info'] = 'pma_table_info';
39  $cfg['Servers'][$i]['table_coords'] = 'pma_table_coords';
40  $cfg['Servers'][$i]['pdf_pages'] = 'pma_pdf_pages';
41  $cfg['Servers'][$i]['column_info'] = 'pma_column_info';
```

Change the values to :

```
/* Authentication type and info */
$cfg['Servers'][$i]['auth_type'] = 'cookie';
$cfg['Servers'][$i]['user'] = 'root';
$cfg['Servers'][$i]['password'] = '12345';
$cfg['Servers'][$i]['extension'] = 'mysqli';
$cfg['Servers'][$i]['AllowNoPassword'] = false;
$cfg['Lang'] = '';
```

password: password is not set in the configuration file.

Testing



The image shows a web browser window with the phpMyAdmin interface. The browser's address bar displays the URL `localhost/phpmyadmin/index.php?route=/`. The phpMyAdmin logo, featuring a sailboat, is centered at the top of the page. Below the logo, the text "Welcome to phpMyAdmin" is displayed. The interface includes a "Language" dropdown menu currently set to "English". Below this is a "Log in" button with a help icon. The login form contains two input fields: "Username:" with the value "root" and "Password:" which is empty. A "Go" button is located at the bottom right of the login form.

phpMyAdmin

localhost/phpmyadmin/index.php?route=/

phpMyAdmin

Welcome to phpMyAdmin

Language

English

Log in

Username: root

Password:

Go

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

